Handling the inversion of a severely ill-conditioned operator: the case of fDOT

Fluorescence diffuse optical tomography (fDOT) for preclinical small-animal imaging

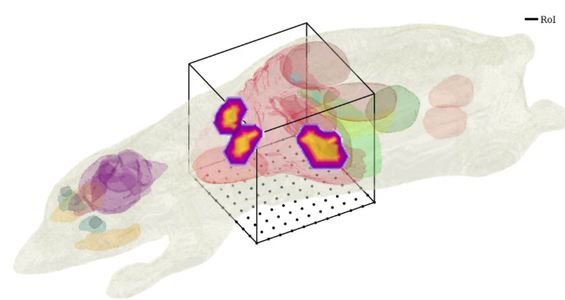
Clément Lavaud Anabela Da Silva Caroline Chaux Martin Rodriguez-Vega Julien Wojak Rémi André Pascal Nguyen

clement.lavaud@univ-amu.fr · Aix-Marseille Université · Institut de Mathématiques de Marseille (I2M) · Institut Fresnel

1. Context: why fDOT for small-animal imaging?

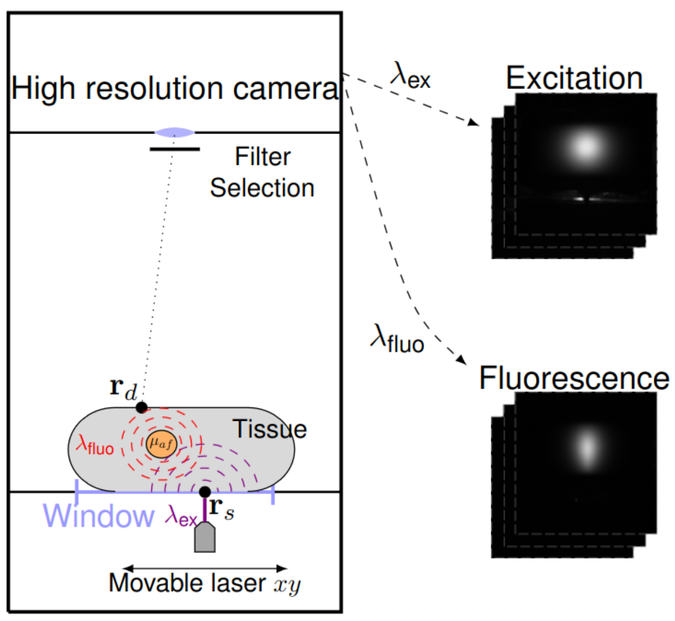
Preclinical imaging uses **small animals**, typically mice, to study tumour development, drug biodistribution and treatment response before clinical use. longitudinal studies on the same animal over time are especially valuable.

- **Molecular contrast.** Near-infrared (NIR) fluorescent probes reveal specific biological processes.
- **Non-ionizing imaging.** Unlike CT or PET, optical imaging uses no ionizing radiation and supports repeated measurements over time.
- **From 2D to 3D.** Planar fluorescence provides limited depth information. **fDOT** reconstructs the fluorophore distribution in 3D, including its location and concentration.
- **The challenge.** Light scatters strongly in tissue, making reconstruction ill-posed and motivating robust reconstruction methods.



Simulated mouse geometry [3] with the region of interest (RoI) and fluorescent inclusions.

2. Acquisition and physics of the problem



Two measurements per source–detector pair: Light emission at excitation and fluorescence wavelengths (respectively λ_{ex} usually $\simeq$ 808 nm and λ_{fluo} usually $\simeq$ 1050 nm).

Physical principle

1. **Excitation.** A NIR-I laser (λ_{ex}) illuminates the animal. We will take several acquisitions corresponding to different source positions $\mathbf{r}_s$.
2. **Fluorescence.** Fluorophores absorb part of the excitation light and re-emit it at a higher wavelength, λ_{fluo}. A filter separates this emission from the excitation light.
3. **Detection.** Two light intensity measurements are recorded by a high-resolution camera placed above the animal: one at the excitation wavelength and one at the fluorescence-emission wavelength.

In highly scattering tissue, light propagation is commonly modelled by the **diffusion equation** [2], which describes the spatial spread and attenuation of excitation and fluorescence light. After discretization, the **forward model** links the fluorophore distribution to the measured signal.

$$\begin{cases} -\nabla\cdot(D_{\mathrm{ex}}\nabla\Phi_{\mathrm{ex}}) + \mu_{a,\mathrm{ex}}\Phi_{\mathrm{ex}} = S, \\ -\nabla\cdot(D_{\mathrm{em}}\nabla\Phi_{\mathrm{em}}) + \mu_{a,\mathrm{em}}\Phi_{\mathrm{em}} = \eta\,\mu_{af}(\mathbf{r})\Phi_{\mathrm{ex}}. \end{cases} \longrightarrow \quad \mathbf{m} = \mathbf{A}\mathbf{c} + \boldsymbol{\varepsilon}, \quad \mathbf{c} \geq 0$$

$\Phi_{\mathrm{ex}}, \Phi_{\mathrm{em}}$: excitation/emission fluence rates.
$D_{\mathrm{ex}}, D_{\mathrm{em}}$: diffusion coefficients.
$\mu_{a,\mathrm{ex}}, \mu_{a,\mathrm{em}}$: background absorption coefficients.
S: excitation source; η: fluorescence quantum yield.
$\mu_{af}(\mathbf{r})$: fluorophore absorption map to reconstruct.

m: stacked camera measurements.
A: discretized forward operator, including light transport and acquisition geometry.
c: nonnegative fluorophore concentration coefficients.
ε: measurement noise and model mismatch.

Why is reconstruction challenging?
Diffusion smooths the signal and suppresses fine spatial details. The discretized model is therefore difficult to invert stably; regularization is essential.

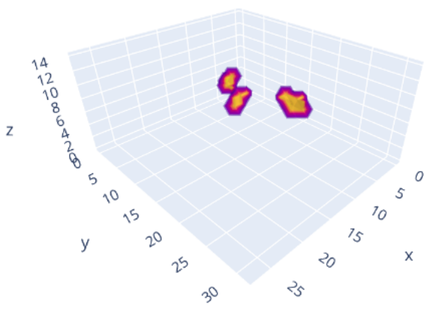
Data
Realistic simulated small-animal database with various geometries for fluorophore inclusions. [3].

3. Least-squares methods and reconstructions

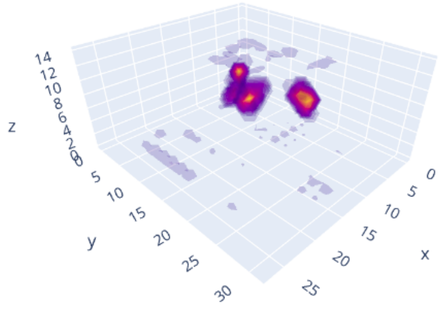
For the Gaussian-noise baseline, assume independent errors with constant variance, $\varepsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I})$. The negative log-likelihood gives the **least-squares (LS)** data term $\frac{1}{2}\|\mathbf{A}\mathbf{c} - \mathbf{m}\|_2^2$. We compare three regularizers $R(\mathbf{c})$ in the same nonnegative Maximum a posteriori (MAP) objective:

$$\hat{\mathbf{c}} \in \arg\min_{\mathbf{c}\geq 0} \left[\frac{1}{2}\|\mathbf{A}\mathbf{c} - \mathbf{m}\|_2^2 + \lambda R(\mathbf{c})\right]$$

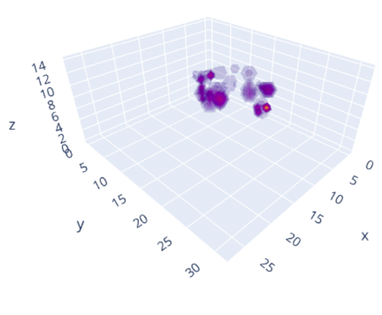
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ground truth "arby1101" [3]



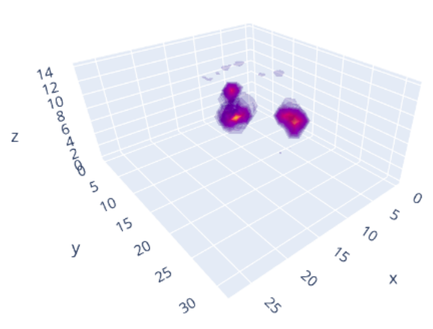
LS + ℓ_2
quadratic penalty
$R(\mathbf{c}) = \frac{1}{2}\|\mathbf{c}\|_2^2$
Adaptive diagonal preconditioning [7]:
$\mathbf{c}^{k+1} = P_+\left[\mathbf{c}^k - \mathbf{D}_k^{-1}\left(\mathbf{A}^\top(\mathbf{A}\mathbf{c}^k - \mathbf{m}) + \lambda_2\mathbf{c}^k\right)\right]$



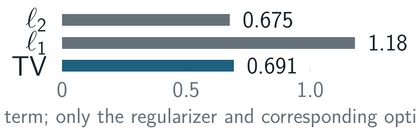
LS + ℓ_1
sparsity penalty
$R(\mathbf{c}) = \|\mathbf{c}\|_1$
Adaptive diagonal proximal gradient [7]:
$\mathbf{c}^{k+1} = P_+\left[\mathbf{c}^k - \mathbf{D}_k^{-1}\left(\mathbf{A}^\top(\mathbf{A}\mathbf{c}^k - \mathbf{m}) + \lambda_1\mathbf{1}\right)\right]$



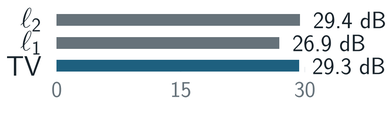
LS + TV
total-variation penalty
$R(\mathbf{c}) = \|\nabla\mathbf{c}\|_{2,1}$
Primal–dual updates (data step + TV dual projection):
$\mathbf{c}^{k+1} = P_+\left[\mathbf{c}^k - \tau\left(\mathbf{A}^\top(\mathbf{A}\mathbf{c}^k - \mathbf{m}) + \nabla^\top\mathbf{q}^k\right)\right]$
$\mathbf{q}^{k+1} = \Pi_{\|\mathbf{q}_i\|_2\leq\lambda_{\mathrm{TV}}}\left(\mathbf{q}^k + \sigma\nabla\bar{\mathbf{c}}^{k+1}\right)$



NMSE
Lower is better



pSNR
Higher is better



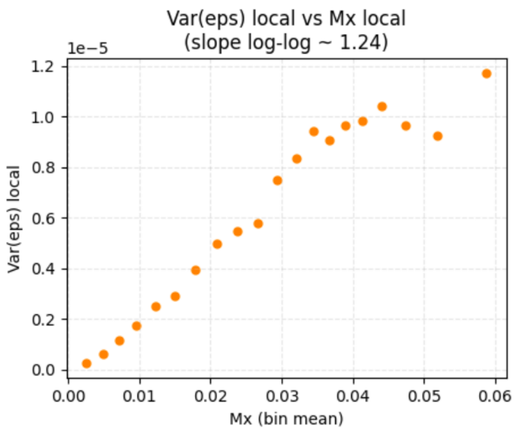
The three reconstructions use the same LS data term; only the regularizer and corresponding optimization update change [4,5].

4. Mean–variance diagnostic and other fidelity terms

Signal-dependent variance

A power-law noise model writes $\mathrm{Var}(M) \propto [\mathbb{E}(M)]^p$. The exponent p describes how quickly variance grows with mean: $p = 0$ is constant variance and $p = 1$ is Poisson-like scaling. A log–log fit estimates p from the data.

The log–log fit gives $p \approx 1.24$, indicating signal-dependent (heteroscedastic) noise. Its variance grows somewhat faster than the Poisson-like case $p = 1$, and unlike constant-variance Gaussian noise ($p = 0$).



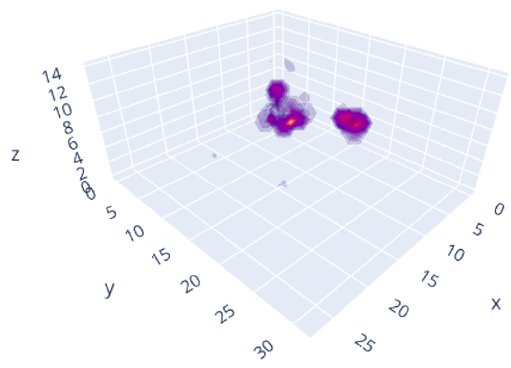
Empirical mean–variance diagnostic (original plot).

Poisson noise ⟶ Kullback-Leibler (KL) fidelity

For independent Poisson measurements, $M_m \sim \mathrm{Poisson}(\mu_m)$ and $\mathrm{Var}(M_m) = \mu_m$. The negative log-likelihood yields a KL data-fidelity term. With TV regularization, we reconstruct $\mathbf{c}$ by minimizing [6]:

$$D_{\mathrm{KL}}(\mathbf{m}\|\mathbf{A}\mathbf{c}) = \sum_m [(\mathbf{A}\mathbf{c})_m - m_m \log((\mathbf{A}\mathbf{c})_m)]$$

$$\bar{\mathbf{c}} = \arg\min_{\mathbf{c}\geq 0} [D_{\mathrm{KL}}(\mathbf{m}\|\mathbf{A}\mathbf{c}) + \lambda_{\mathrm{TV}}\|\nabla\mathbf{c}\|_{2,1}]$$



KL + TV reconstruction

Open question The measured scaling $p \approx 1.24$ is not captured by the constant-variance Gaussian ($p = 0$) or Poisson ($p = 1$) models. Can the **data-fidelity term** be learned from the observed error statistics?

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[7] Y. Park, S. Dhar and S. Boyd, "Variable Metric Proximal Gradient Method with Diagonal Barzilai–Borwein Stepsize," arXiv:1910.07056, 2019.